

RUG POOL

A Provably Fair Memecoin Game on Monad

The rug is guaranteed. The question is whether it hits you.

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Built on Monad | Supported by DeltaV

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1. Abstract

Rug Pool is a provably fair memecoin launch and trading game built on Monad. Every coin launched on the platform locks all holders for 24 hours. At the 24-hour mark, Monad's onchain verifiable random function (VRF) determines the outcome for every holder simultaneously.

The mechanic uses a configurable flip probability. In the default 33.3/66.6 configuration, approximately 33% of holders exit each cycle (Heads) and 66% survive to the next (Tails). Those who exit are removed from the game. Those who flip Tails sacrifice their full current position value to the pool — funding cleaner exits for winners — and remain locked in the next cycle with a ghost token position that can recover if new buyers enter.

The exit queue is ordered by cycle age first, then join time within the same cycle. Survivors who have been locked longer always exit before newer buyers. This rewards commitment and penalises late entry.

10% of all protocol fees accumulated that month across every coin and every cycle goes to one single top loser — the wallet with the largest verified loss across the entire protocol. Losing big is not just a risk — it is a legitimate strategy.

2. The Problem

2.1 Memecoin Launches Are Rigged

In every standard memecoin launch, insiders buy at the earliest price, dump on retail at the peak, and retail holders are left with bags and no recourse. The pattern is so consistent it has become an expectation rather than an exception. The core issue is asymmetric exit timing. Insiders know when the rug is coming. Retail does not.

2.2 Losing Has No Upside

When a memecoin rugs, losers leave with nothing but a lesson. There is no mechanism in any existing memecoin platform that rewards the downside. The worst outcome is purely punitive. This destroys retention — people who lose stop playing.

2.3 Bots and Sybils Destroy Fair Launches

Free-to-join platforms are immediately flooded with bots, sybil wallets, and fake volume. This distorts price discovery, inflates holder counts, and degrades the experience for real users playing with real money.

2.4 No Native Game for the Monad Meme Community

Monad has 10,000 TPS and 0.8 second finality — infrastructure purpose-built for real-time applications. No existing memecoin platform is built specifically for this community or takes advantage of this performance. The community exists. The game does not.

3. The Solution

Problem	Solution
Rigged launches	24-hour lock applies to everyone equally. No whitelist exits. No dev sells. Not even the creator can exit early.
Losing has no upside	10% of monthly protocol fees go to the top loser. Worst loss of the month wins a prize.
Bots and sybils	\$1 non-refundable founding member fee. Proof of human. Bots cannot justify \$1 per wallet at scale.
No native Monad game	Built from the ground up on Monad for the Monad meme community.

4. How It Works

4.1 The Global Timer

Every coin has a single global countdown timer. When a coin launches, the 24-hour clock starts for everyone. Every holder flips simultaneously when the timer hits zero — regardless of when they bought within the cycle.

There is no per-wallet timer. One coin. One clock. One flip. The shared simultaneous moment is what drives community activity, live chat, and social media virality. Everyone finds out at the same time.

4.2 The Flip Mechanic

At T+24, Monad's onchain VRF generates one random seed per coin per cycle. Each holder's outcome is derived deterministically from this shared seed:

```
seed = VRF.requestRandomness()    // one call per coin per cycle

for each holder:
    hash      = keccak256(seed + walletAddress + cycleNumber)
    outcome = uint256(hash) % 100

    if outcome < exitProbability:    // e.g. 33 for 33.3/66.6 config
        -> Heads (enter exit queue)
    else:
```

-> Tails (sacrifice + locked next cycle)

Every outcome is provably fair and independently verifiable by anyone onchain.

4.3 The Exit Queue — Cycle Age Priority

Holders who flip Heads exit in a queue ordered by two factors:

- Cycle age — holders who have been locked the longest exit first
- Join time within the same cycle age — earlier buyers exit before later buyers in the same tier

Example queue at T+48 (Cycle 2 flip):

Tier 1: Cycle 2 survivors (locked since Cycle 1)
-> Bob (joined Cycle 1 at T=2hrs) exits 1st
-> Carol (joined Cycle 1 at T=8hrs) exits 2nd

Tier 2: New Cycle 2 buyers
-> Dave (joined Cycle 2 at T+24) exits 3rd
-> Emma (joined Cycle 2 at T+26) exits 4th

Exits spaced 10 seconds apart to distribute price impact.

Survivors always exit before new buyers. This is the reward for surviving a Tails flip and staying locked through another cycle. New buyers entering a later cycle accept that they are behind survivors in the queue — that is the known risk of late entry.

4.4 The Sacrifice Mechanic

Holders who flip Tails do not simply stay locked. Their full current position value is sacrificed to the pool before winners extract. This funds cleaner exits for Heads holders:

STEP 1: Calculate all holders current token values at flip time

STEP 2: Tails holders sacrifice full current value to pool
Pool grows by total sacrifice amount
Tails holders tokens remain (ghost position)
Ghost token value = near zero after sacrifice

STEP 3: Heads holders exit in cycle age queue order
Each winner: 80% of current token value to wallet

20% of current token value to treasury

STEP 4: Cycle N+1 begins

Survivors locked with ghost positions

New buyers can enter and pump price

Ghost positions recover via bonding curve as price rises

4.5 Ghost Position Recovery

When a Tails holder sacrifices, their tokens are not burned — they remain onchain. The token value drops to near zero because the pool drained. This is a ghost position.

Recovery happens naturally through the bonding curve. Every new buyer in the next cycle adds MON to the pool and mints new tokens, pushing the price up. As price rises, the survivor's ghost token count is worth more. The survivor does not need special tracking — the bonding curve handles it automatically.

```
Bob sacrificed at Cycle 1 end:  
Tokens held:  1,000,000 MPEPE  
Value:        ~$0 (pool drained)
```

```
Cycle 2 — new buyers enter:  
Dave buys $20  -> price pumps  
Emma buys $15  -> price pumps again
```

```
Bob's ghost position recovers:  
Same 1,000,000 MPEPE tokens  
New value: ~$8 (price recovered via new liquidity)
```

```
Bob's only other option: skill MPEPE hard and pray.  :
```

4.6 Settlement Order When Both Tiers Flip Heads

If Cycle 2 survivors AND new Cycle 2 buyers both flip Heads at T+48, settlement follows strict priority:

- Tails sacrifices happen first — all Tails holders sacrifice current value to pool
- Tier 1 (cycle age survivors) exit first — they drain first, best prices
- Tier 2 (new cycle buyers) exit after — they exit into a pool already reduced by Tier 1

New buyers entering a later cycle accept this risk explicitly. The cycle age of every survivor is visible onchain before anyone buys in.

4.7 Re-entry

Any holder who has been auto-sold can re-enter any coin at any time. Re-entry places them at the back of the exit queue for the current cycle at Tier 2 position. They flip with everyone else at the next global T+24. There is no advantage to timing re-entry within a cycle — the flip hits everyone simultaneously.

5. Worked Example — Two Full Cycles

Cycle 1 Setup

```
Alice buys $10 of MPEPE at T=0      (earliest, lowest price)
Bob   buys $10 of MPEPE at T=2hrs
Carol buys $10 of MPEPE at T=8hrs   (latest, highest price)
```

```
Flip config: 33.3% exit, 66.6% survive
Total pool: $30
```

Buying activity pumps price via bonding curve:

```
Alice tokens now worth: $14 (bought the bottom)
Bob   tokens now worth: $12
Carol tokens now worth: $10 (bought the top, least gain)
Total pool value at T+24: $36
```

T+24 — Cycle 1 Flip

```
Alice -> Heads  (exit queue, position 1)
Bob   -> Tails  (sacrifice, locked Cycle 2)
Carol -> Tails  (sacrifice, locked Cycle 2)
```

Step 1 — Bob and Carol Sacrifice

```
Bob sacrifices:    $12 (tokens remain as ghost position)
Carol sacrifices:  $10 (tokens remain as ghost position)
Total sacrifice:   $22
```

Pool before: \$36
Pool after: \$58

Step 2 — Alice Exits

Alice current value: \$14
Alice receives: 80% of \$14 = \$11.20 profit +\$1.20
Treasury gets: 20% of \$14 = \$2.80
Pool after: \$44

Cycle 1 Outcomes

Player	Net P&L	Status
Alice	+\$1.20	Bought early, only winner, best outcome
Bob	-\$12 (ghost)	Tails, sacrificed, enters Cycle 2 with ghost tokens
Carol	-\$10 (ghost)	Tails, sacrificed, enters Cycle 2 with ghost tokens
Treasury	+\$2.80	Accumulates toward monthly payout

Cycle 2 — Ghost Recovery

Pool entering Cycle 2: \$44
Bob ghost tokens: worth ~\$0
Carol ghost tokens: worth ~\$0

New buyers enter:
Dave buys \$20 at T+24 -> price pumps
Emma buys \$15 at T+26 -> price pumps again

Pool grows: $\$44 + \$20 + \$15 = \79

Ghost positions recover via bonding curve:
Bob tokens now worth: ~\$8
Carol tokens now worth: ~\$6

T+48 — Cycle 2 Flip

```
Bob   -> Heads   (Tier 1, cycle age survivor, exits 1st)
Carol -> Heads   (Tier 1, cycle age survivor, exits 2nd)
Dave  -> Tails   (Tier 2, new buyer, sacrifice)
Emma  -> Tails   (Tier 2, new buyer, sacrifice)
```

Cycle 2 Settlement

```
Dave and Emma sacrifice first:
  Dave:  current value $20 -> sacrificed to pool
  Emma:  current value $15 -> sacrificed to pool
Pool before:  $79
Pool after:   $79 + $35 = $114
```

```
Bob exits (Tier 1, position 1):
  Current value: $8
  Bob receives:  80% of $8 = $6.40
  Treasury:      20% of $8 = $1.60
  Pool after:    $104.40
```

```
Carol exits (Tier 1, position 2):
  Current value: $6
  Carol receives: 80% of $6 = $4.80
  Treasury:       20% of $6 = $1.20
  Pool after:     $98.40
```

```
Dave and Carol enter Cycle 3 as ghost positions.
Pool: $98.40 waiting for new buyers.
```

Full Two-Cycle Outcomes

Player	Net P&L	Notes
Alice	+\$1.20	Cycle 1 winner, exited clean
Bob	-\$3.60	Sacrificed \$12, recovered to \$8, exited at \$6.40
Carol	-\$5.20	Sacrificed \$10, recovered to \$6, exited at \$4.80
Dave	-\$20.00	New Cycle 2 buyer, sacrificed full position, Cycle 3 ghost
Emma	-\$15.00	New Cycle 2 buyer, sacrificed full position, Cycle 3 ghost

Treasury	+\$5.60	\$2.80 + \$1.60 + \$1.20, accumulates monthly
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6. Token Economics

6.1 Flow of Value

Flow	Allocation
Auto-sell exit value	80% to holder, 20% to protocol treasury
Protocol treasury (monthly)	10% to dev wallet, 10% to single protocol-wide top loser, 80% rolls over
Founding member fee	\$1 per user, non-refundable, one-time
Verified project badge	\$10 per coin launch, priority feed placement

6.2 The Protocol Treasury

20% of every auto-sell exit value across all coins and all cycles accumulates in the protocol treasury. The treasury does not pay out per exit. It accumulates continuously and distributes monthly:

- 10% to the dev wallet — sustains protocol development
- 10% to the single protocol-wide top loser — one winner per month, the wallet with the largest verified loss across all coins and all cycles
- 80% rolls over to the following month — compounds over time, grows the top loser prize

6.3 The Bonding Curve

Every coin uses an automated market maker bonding curve. Every buy increases the price. Every sell decreases it.

$$\text{Price} = \text{Total Pool Value} / \text{Total Token Supply}$$

On buy: MON added to pool, new tokens minted -> price increases

On sell: MON removed from pool, tokens burned -> price decreases

Consequence:

Early buyers	-> lowest price, most tokens, biggest gains if price pumps
Late buyers	-> highest price, fewest tokens, least gain, last in queue
Every buy	-> pumps price for all holders who bought before

6.4 The Founding Member Fee

The first 1,000 users pay a one-time \$1 non-refundable fee. This is proof of human, not a platform charge. The fee pool is allocated:

- \$500 seeds the initial top loser prize pool — real money in the leaderboard from day one
- \$500 to the dev — covers initial infrastructure costs

7. Flip Configurations

7.1 Available Configurations

Config	Mode	Character
33.3 / 66.6	Default	1 in 3 exits each cycle. Majority survive and sacrifice. High drama, slow grind.
50 / 50	Classic	Half exit each cycle. Fast death, high volatility, extreme risk.
70 / 30	Extended	Most exit, few survive. Coins die faster but winners win bigger.
80 / 20	Marathon	Very slow exit. Maximum survivor count. Low volatility, long game.

7.2 Cycle Survival Comparison

Starting with 100 holders:

	33/66	50/50	70/30	80/20	
Cycle 1:	67	50	30	20	survive
Cycle 2:	45	25	21	16	
Cycle 3:	30	13	15	13	
Cycle 4:	20	6	10	10	
Cycle 5:	13	3	7	8	
Cycle 6:	9	1	5	7	

33/66 keeps coins alive longest with meaningful holder counts.
Coins still active at Cycle 6+ create week-long narratives.

The flip configuration is set at launch and cannot be changed mid-coin. It is displayed on every coin card so buyers can assess risk before entering.

8. Badge System

Badge	Cost	Meaning
Blue	\$1	Verified human. Paid the founding member fee. Any user who registered on the platform.
Yellow	\$10	Verified project. Reviewed for legitimacy. Priority placement at top of coin feed.

Yellow verified projects receive priority placement at the top of the feed. The fee is not for the checkmark — it is for visibility in front of active Rug Pool users. Badges are onchain and non-transferable.

9. Platform Dashboard

Stat	Definition
Total Pool	Combined protocol treasury value across all active coins
Active Coins	Coins currently live with holders locked in
Total Holders	Every unique wallet currently locked — live count
24h Volume	Total buy volume across all coins in the last 24 hours
Total Rugged	Cumulative auto-sell value since launch — only goes up (Elmo fire background)
Survivors	Wallets currently holding Tails — locked right now (Hercules smirk background)

10. Contract Architecture

Contract	Purpose
MemberRegistry.sol	Founding member fee collection, badge issuance, human verification
CoinFactory.sol	Coin launch, bonding curve, token creation and management
RugPool.sol	Game lifecycle, global timer, 24hr lock, flip trigger, cycle age tracking, exit queue
VRFConsumer.sol	Pyth Entropy integration — one seed per coin per cycle
Treasury.sol	Pool accumulation, monthly dev payout, top loser prize distribution

10.1 Cycle Age Tracking

The RugPool contract tracks two values per holder per coin:

- cycleJoined — the cycle number when the holder first bought in
- joinTimestamp — the block timestamp of their first buy in that cycle

```
struct HolderInfo {
    uint256 cycleJoined;      // cycle age - lower = older = higher priority
    uint256 joinTimestamp;    // tiebreaker within same cycle age
    uint256 tokenBalance;     // current token count including ghost positions
}

// Exit queue sort:
// Primary:   ascending cycleJoined  (older cycle = exits first)
// Secondary: ascending joinTimestamp (earlier join = exits first)
```

10.2 VRF Implementation

One VRF call per coin per cycle. All holder outcomes derived from the shared seed:

```

uint256 seed = PythEntropy.requestRandomness();

for each holder {
    bytes32 hash = keccak256(abi.encodePacked(
        seed, holderAddress, cycleNumber
    ));
    uint256 outcome = uint256(hash) % 100;
    bool isHeads = outcome < exitProbability;
}

// Batch processed: 1,000 wallets per transaction
// Gas cost per cycle: under $10 for 200,000 users on Monad

```

11. Technical Stack

Layer	Technology
Blockchain	Monad — EVM-compatible, 10,000 TPS, 0.8s finality
Randomness	Pyth Entropy VRF — provably fair, manipulation-proof
Smart Contracts	Solidity 0.8.28 compiled with Foundry
Frontend	Svelte 5 + Vite — 7 pages, 17 components
Wallet Auth	Privy — embedded wallets, Google sign-in support
Backend	Node.js + Express + viem — event indexer and API layer
Real-time	WebSocket — live feed, countdowns, chat, flip broadcasts
Charts	lightweight-charts v5 — per-coin price charts

12. Build Architecture — Inside Out

Rug Pool is built layer by layer from the inside out. Each layer is fully tested before the next is built. Bugs surface early when they are cheap to fix.

Layer	Purpose
Layer 1 — Contracts	All 5 Solidity contracts with full logic
Layer 2 — JS test scripts	One script per contract hitting deployed testnet
Layer 2.5 — Simulation script	5-minute cycle simulation for rapid testing
Layer 3 — Chain script	Combines all scripts into a full game cycle test
Layer 4 — API	Express server exposing chain script functions as endpoints
Layer 5 — Frontend	Svelte frontend consuming the API — no mock data

13. Roadmap

Phase 1 — Build (72hrs)

- All 5 contracts with full logic
- JS test scripts per contract
- Chain simulation script — full game cycle in 5 minutes
- Backend indexer, WebSocket server, API layer
- Frontend wired to API, all mock data removed

Phase 2 — Testnet

- Deploy all contracts to Monad testnet
- 1,000 user public testnet with live flip events
- Community feedback and iteration
- Monad ecosystem grant application with testnet traction data

Phase 3 — Mainnet

- Security review and audit
- Mainnet deployment
- \$1 founding member onboarding — 1,000 spots
- First live simultaneous flip event
- KOL partnerships and community growth

Phase 4 — Scale

- Queue-based exit as optional coin launch mode
 - Configurable sacrifice percentage (partial vs full)
 - Mobile app
 - Governance for yellow badge verification
 - Cross-chain expansion
-

14. Risks and Disclaimers

14.1 Financial Risk

Rug Pool involves real money and real risk. Tails holders sacrifice their full current position value. The protocol does not guarantee any return. Ghost positions may not recover if no new buyers enter the next cycle.

14.2 Smart Contract Risk

Smart contracts may contain bugs or vulnerabilities. Rug Pool will undergo a security review before mainnet launch. Audit status will be published before any real funds are accepted on mainnet.

14.3 VRF Dependency

The flip mechanic depends on Pyth Entropy for randomness. Any disruption to Pyth Entropy on Monad could delay flip execution. Fallback mechanisms will be documented before mainnet launch.

14.4 Regulatory

Rug Pool is a game. The platform does not provide financial advice. Users are responsible for understanding the legal status of participation in their jurisdiction.

15. Conclusion

Rug Pool does not pretend the rug is not coming. It puts it on a schedule, makes it equal for everyone, turns the worst outcome into a competition, and gives survivors a fighting chance if they earn it.

The mechanics are simple enough to explain in two sentences but deep enough to generate real strategy. Early buyers earn priority exit. Survivors earn cycle age advantage. Losers earn a shot at the monthly prize. The dev is locked in too.

Built on Monad because nothing else is fast enough. Launched in 72 hours because the best proof is a working product.

Memes on Monad.

Rugs, pumps, holds, dumps — all scheduled.

Buy and pray.